(for example, Cm^{3+} is extracted easier, i.e. at a lower pH, than Am^{3+}, etc.), as the ionic radii decrease in that order.

The chemistry involved in the isolation and purification of the actinide elements from irradiated reactor fuel elements is further discussed in Chapter 21. Actinide chemistry in the ecosphere is discussed in §22.6.

14.4. USES OF ACTINIDES

The use of the actinide elements fall into three categories: (i) for understanding fundamental chemistry and the nature of the periodic system, (ii) as fuels and products in the large scale use of nuclear energy, and (iii) miscellaneous applications, where the particular physical, chemical or nuclear properties are valuable. Only the last aspect is discussed here, the others are treated elsewhere in this book. The availability of transuranium element isotopes suitable for experiments is listed in Table 14.6.

Table 14.6 Availability of transuranium element materials

Nuclide	$t_{½}$	Decay mode	Specific amounts[†] available	Specific activity (Bq/g)
^{237}Np	2.14×10^6 y	α, SF(10^{-10} %)	kg	2.61×10^7
^{238}Pu	87.7 y	α, SF(10^{-7} %)	kg	6.33×10^{11}
^{239}Pu	2.41×10^4 y	α, SF(10^{-4} %)	kg	2.30×10^9
^{240}Pu	6.56×10^3 y	α, SF(10^{-6} %)	kg	8.40×10^{10}
^{241}Pu	14.4 y	β, α(10^{-3} %)	1–10 g	3.82×10^{12}
^{242}Pu	3.76×10^5 y	α, SF(10^{-3} %)	100 g	1.46×10^8
^{244}Pu	8.00×10^7 y	α, SF(10^{-1} %)	10–100 mg	6.52×10^5
^{241}Am	433 y	α , SF(10^{-10} %)	kg	1.27×10^{11}
^{243}Am	7.38×10^3 y	α, SF(10^{-8} %)	10–100 g	7.33×10^9
^{242}Cm	162.9 d	α, SF(10^{-5} %)	100 g	1.23×10^{14}
^{243}Cm	28.5 y	α,ε (0.2 %)	10–100 mg	1.92×10^{12}
^{244}Cm	18.1 y	α, SF(10^{-4} %)	10–100 g	3.00×10^{12}
^{248}Cm	3.40×10^5 y	α, SF(8.3 %)	10–100 mg	1.57×10^8
^{249}Bk	320 d	β,α(10^{-3} %), SF(10^{-8} %)	10–50 mg	6.00×10^{10}
^{249}Cf	350.6 y	α, SF(10^{-7} %)	1–10 mg	1.52×10^{11}
^{250}Cf	13.1 y	α, SF(0.08 %)	10 mg	4.00×10^{12}
^{252}Cf	2.6 y	α, SF(3.1 %)	10–1000 mg	2.00×10^{13}
^{254}Cf	60.5 d	SF, α (0.3 %)	μg	3.17×10^{14}
^{253}Es	20.4 d	α, SF(10^{-5} %)	1–10 mg	9.33×10^{14}
^{254}Es	276 d	α	1–5 μg	6.83×10^{13}
^{257}Fm	100.5 d	α, SF(0.2 %)	1 pg	1.83×10^{14}

[†]According to Oak Ridge Nat. Lab., USA, late 1990:ies.

Table 14.7 New and historic names of the transactinide elements

Atomic number	IUPAC name	Symbol	Old IUPAC name	Old suggested name
104	rutherfordium	Rf	Db	Rf
105	dubnium	Db	Joliotium (Jl)	Hahnium (Ha)
106	seaborgium	Sg	Rf	Sg
107	bohrium	Bh	Bh	Nielsbohrium (Ns)
108	hassium	Ha	Hahnium (Hn)	Hs
109	meitnerium	Mt	Mt	Mt
110	darmstadtium	Ds		
111	roetgenium	Rg		
112	copernicium	Cn		
113				
114	flerovium	Fl		
115				
116	livermorium	Lv		

Table 14.8 Nuclear reactions used for the first synthesis of elements 104–116

Element	Discovery (year)	Discovery reaction
104	1969	$^{249}_{98}\mathrm{Cf} + {}^{12}_{6}\mathrm{C} \rightarrow {}^{257}_{104}\mathrm{Rf} + 4\mathrm{n}$
105	1970	$^{249}_{98}\mathrm{Cf} + {}^{15}_{7}\mathrm{N} \rightarrow {}^{260}_{105}\mathrm{Db} + 4\mathrm{n}$
106	1974	$^{249}_{98}\mathrm{Cf} + {}^{18}_{8}\mathrm{O} \rightarrow {}^{263}_{106}\mathrm{Sg} + 4\mathrm{n}$
107	1981	$^{209}_{83}\mathrm{Bi} + {}^{54}_{24}\mathrm{Cr} \rightarrow {}^{262}_{107}\mathrm{Bh} + \mathrm{n}$
108	1984	$^{208}_{82}\mathrm{Pb} + {}^{58}_{26}\mathrm{Fe} \rightarrow {}^{265}_{108}\mathrm{Hs} + \mathrm{n}$
109	1982	$^{209}_{83}\mathrm{Bi} + {}^{58}_{26}\mathrm{Fe} \rightarrow {}^{266}_{109}\mathrm{Mt} + \mathrm{n}$
110	1994	$^{208}_{82}\mathrm{Pb} + {}^{62}_{28}\mathrm{Ni} \rightarrow {}^{269}_{110}\mathrm{Ds} + \mathrm{n}$
111	1994	$^{209}_{83}\mathrm{Bi} + {}^{64}_{28}\mathrm{Ni} \rightarrow {}^{272}_{111}\mathrm{Rg} + \mathrm{n}$
112	2001	$^{208}_{82}\mathrm{Pb} + {}^{70}_{30}\mathrm{Zn} \rightarrow {}^{277}_{112}\mathrm{Cn} + \mathrm{n}$
113	2004	$^{243}_{95}\mathrm{Am} + {}^{48}_{20}\mathrm{Ca} \rightarrow {}^{288}_{115}\mathrm{Uup} + 3\mathrm{n} \rightarrow {}^{284}_{113}\mathrm{Uut} + \alpha$
114	2002	$^{244}_{94}\mathrm{Pu} + {}^{48}_{20}\mathrm{Ca} \rightarrow {}^{289}_{114}\mathrm{Fl} + 3\mathrm{n}$
115	2004	$^{243}_{95}\mathrm{Am} + {}^{48}_{20}\mathrm{Ca} \rightarrow {}^{288}_{115}\mathrm{Uup} + 3\mathrm{n}$
116	1961	$^{248}_{96}\mathrm{Cm} + {}^{48}_{20}\mathrm{Ca} \rightarrow {}^{293}_{116}\mathrm{Lv} + 3\mathrm{n}$

The spontaneous decay of the transuranium elements by α-decay and/or fission results in energy release. Since also very small amounts of some nuclides (e.g. ^{238}Pu, ^{244}Cm, ^{252}Cf) can be sources of appreciable energy, these radionuclides can be used in *small power generators*. This use of actinide elements is discussed in §7.9.3.

^{241}Am emits a 60 keV γ-ray and has been used as a γ-radiation source to measure thickness of metal sheets and of deposited metal coatings, the degree of soil

compaction, sediment concentration in flowing streams, and to induce X-ray fluorescence in chemical analysis (§7.8). As an α-particle emitter, ^{241}Am has been mixed with beryllium to make intense neutron sources for oil well logging and for measuring water content in soils and in process streams in industrial plants. It is extensively used for elimination of static electricity and in smoke detectors where its use depends on the ionization of air.

^{252}Cf ($t_{½}$ 2.73 y) has found several uses. It decays to 3.1% by spontaneous fission (the main decay mode is through α emission) leading to a neutron emission rate of 2.3×10^{15} n s^{-1} kg^{-1} (average n-energy 2.35 MeV). The n-dose rate is 22 kSv h^{-1} kg^{-1}. ^{252}Cf is the only nuclide that can provide a useful neutron intensity over a sufficiently long half-life to make it a useful single nuclide neutron source. The low rates of heat emission (38.5 kW/kg), γ-radiation (initially 1.3×10^{16} photons s^{-1} kg^{-1}, yielding an initial dose rate of 1.6 kGy h^{-1} kg^{-1}; γ-ray intensity increases with time due to fission product build-up) and helium evolution (from α decay) allow fabrication of simple, small ^{252}Cf sources that require no external power supply nor any maintenance but that can provide moderately high neutron fluxes. The 2003 production rate of ^{252}Cf from the Oak Ridge facilities (HFIR-TRU) was ~ 250 mg/y and from russian Dimitrograd ~25 mg/y. Among the applications of ^{252}Cf we list the following:

(a) process control by a variety of on-stream non-destructive analytical techniques;
(b) medical diagnosis by activation analysis;
(c) production of short-lived radioisotopes at locations where they will be used, thus avoiding decay during transportation from an accelerator or reactor at another site;
(d) industrial neutron radiography, which images low density materials — especially hydrogenous materials — better than X-rays;
(e) possible medical treatment of tumors with ^{252}Cf sources that can be implanted in the body;
(f) petroleum and mineral exploration in which the compactness and portability of ^{252}Cf neutron sources facilitate testing for valuable deposits, particularly in inaccessible places such as deep wells and the sea floor;
(g) moisture measurements;
(h) hydrology studies to locate sources of water;
(i) nuclear safeguards tests, e.g. for criticality control in reactor fuel storage areas and for nuclear materials accountability (detection and recovery of fissionable material; implementation of nuclear agreements).

14.5. TRANSACTINIDE PROPERTIES

The transactinide elements (sometimes also called Super Heavy Elements) have only been produced a few atoms at a time, When the number of atoms observed in an experiment is very small the usual fundamental thermodynamic descriptions no longer